Lê Quý Toàn

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CAREER OBJECTIVE

Detail-oriented, soon-to-graduate Information Technology student with practical experience in Full-stack Web development (React, TypeScript) and a strong passion for Interactive Web Game/Playable Ads development. Possess deep knowledge of OOP, logical thinking, and 3D/Particle graphics algorithms. As a proactive learner, I consistently leverage AI tools daily to accelerate debugging and optimize performance. Currently seeking a Playable Ads / Game Developer Intern position to build high-performance interactive ad experiences and grow in a creative team environment.

EDUCATION

Post and Telecomunications Institute of Technology (PTIT) Oct 2021 - Now

Information Technology Engineer

SKILLS

Programming Languages & Frameworks:	C#, TypeScript, JavaScript, HTML, CSS, React, Redux Toolkit, Vite, OOP & Design Patterns.
Game Engines & Libraries:	Cocos Creator (Basic/Self-study), Phaser, Unity Luna (Concept), Particle-based 3D Algorithms.
Databases & Cloud:	SQL Server, MySQL, Firebase, Supabase, MongoDB, Redis
AI & Advanced Tech:	Gemini API, RAG Architecture, AI-Assisted Development Tools
Other:	Git, GitHub, API Integration, AI-Assisted Development Tools
Soft skills:	Proactive in learning new technologies via AI, Fast adaptability, Problem-solving

WORK EXPERIENCE

MobiFone Information Technology Center June 2025 - Sep 2025

IT Intern

- Familiarized with real-world software development processes and corporate workplace culture.
- Analyzed source code, technical documentation, and system architectures of ongoing projects.
- Received direct mentorship and executed foundational programming tasks (e.g., HTML/CSS integration, building UI components, and developing small-scale APIs).
- Acquired practical experience with version control tools (Git/GitHub) and collaborative teamwork.

PROJECTS

Chicken Invaders - 2D Space Shooter & Playable Prototype May 2026 - June 2026

Description: Developed a 2D top-down space shooter game inspired by the classic Chicken Invaders, focusing on building modular gameplay systems, automated workflows, and smooth player interactions suitable for interactive ad experiences.
Team size: 1
Position in project: Unity / Gameplay Programmer
Technology description: Unity Engine, C#, New Input System, Custom Editor Scripting, Coroutines, Particle Systems, AI-Assisted Debugging Tools.

Responsibility in project:

- Designed and implemented core gameplay mechanics, including responsive spaceship movement and multi-touch controls using Unity's New Input System.
- Developed a custom Flock Management & Grid Logic system for synchronized enemy formation, automated boundary detection, and randomized projectile drops.
- Built a dynamic weapon upgrade system (supporting 1 to 5 fan-shaped laser spreads) integrated with a percentage-based loot drop algorithm.
- Programmed game loop state systems, leveraging Coroutines to handle complex real-time behaviors (player invincibility state, flashing visual feedback, and infinite scrolling backgrounds).
- Wrote Custom Unity Editor Tools (C#) to automate scene configuration, streamlining the generation of explosion particle prefabs and auto-correcting axis orientation for imported 3D models (.glb) into the 2D workspace.
- Leveraged AI coding assistants to accelerate the optimization of math-heavy vector physics and clean up modular C# code architectures.

Game Hub - Lightweight Multiplayer Web Games Portal Jan 2026 - Mar 2026

Description: Developed a high-performance, cross-platform web gaming portal from scratch featuring 6 distinct 2D games (Tetris, Wood Block Puzzle, Snake, Caro, etc.). Focused on custom engine architecture, lightweight rendering, and responsive mobile controls tailored for interactive web environments.

Team size: 1

Position in project: Game Developer / Gameplay Engineer

Technology description: HTML5 Canvas API, Vanilla JavaScript (ES6+), GSAP (Tweening), Node.js, Socket.io (WebSockets), Firebase, AI-Assisted Optimization.

Responsibility in project:

- Built a custom lightweight 2D Web Game Engine executing a core game loop via requestAnimationFrame with Delta Time calculation to guarantee a locked 60 FPS across diverse mobile browsers.
- Optimized graphic rendering pipelines using HTML5 Canvas API, minimizing Draw Calls and managing asset memory to drastically reduce initial loading time (crucial for playable ad retention).
- Integrated GSAP for high-performance UI tweening animations and engineered a custom programmatic Particle Effect system for dynamic in-game feedback.
- Implemented a Finite State Machine (FSM) pattern to handle modular game flows (Menu, Playing, Paused, Game Over) across all 6 mini-games, ensuring clean and maintainable code architecture.
- Designed a cross-platform Input Controller module supporting both desktop keyboard layouts and fluid, mobile-first multi-touch/swipe gestures.
- Programmed complex gameplay math, including 2D Grid Coordinate systems, matrix rotations for puzzle blocks, and accurate collision detection algorithms.
- Developed a Heuristic Evaluation AI for a 15x15 Caro (Gomoku) game with scalable difficulty levels, alongside probability-based AI dealer simulations for card games.
- Architected a real-time multiplayer network using Node.js and Socket.io, implementing room generation/matchmaking systems and optimizing client-server Game State Synchronization to achieve sub-millisecond latency.

INTERESTS

Active individual with a strong passion for playing sports and participating in dynamic group activities. Enthusiastic about the gaming industry, frequently exploring newly released titles, and actively engaging with diverse gaming content across multiple platforms and devices.

CERTIFICATIONS

JavaScript Algorithms and Data Structures – freeCodeCamp	2025
Aptis ESOL International Certificate (CEFR Level B2) – British Council	2026-2028